a Nitrogenase family tree, with traits

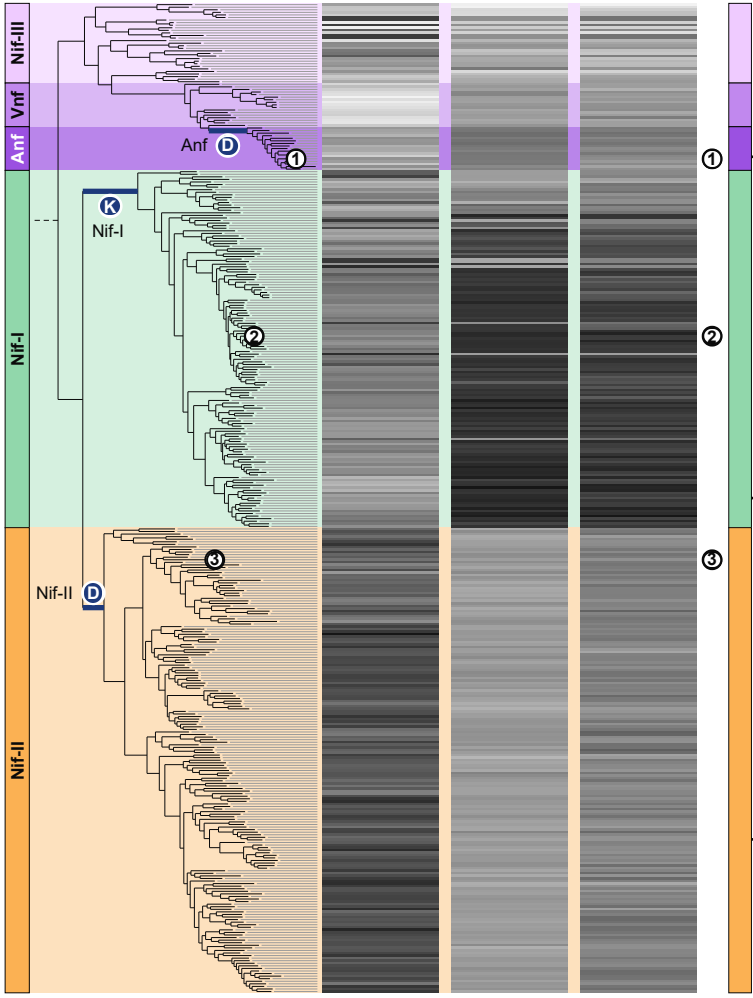
385 living proteins, 384 reconstructed ancestors. Darker bands: Vnf and Anf inside Nif-III.
Nif, Vnf, Anf: nitrogenases using molybdenum, vanadium, or only iron. I to III: Nif clades.
Anf sits inside Vnf. D, K: subunits of the enzyme core. Branch lengths are to scale.

Predicted core (2 D, 2 K)

AlphaFold2 predictions, one row per living protein
Lighter to darker: smaller to larger (area, contacts).
ΔG, predicted binding energy: less to more stable.

Dashed stub (left edge of tree): root branch to 385 outgroup sequences (not drawn).

Surface area (Å²)	Contacts (residue pairs)	ΔG (kcal/mol)
57,616 68,516	703 1,273	−52.6 −101.8



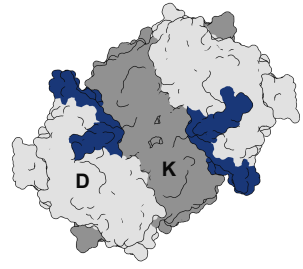
0.5 amino-acid substitutions per site

b Three insertions in deposited structures

The enzyme core holds two D and two K subunits.
Pictures: X-ray entries of the Protein Data Bank (PDB).
D subunit K subunit inserted segment

① Ring and number: the tip whose entry is shown below.
D D or K on a thick blue branch: where the paper places that insertion; the clade is named beside it. K appears first, then lengthens later.
Residues: computed here; "Paper says": the paper's value.

① 8BOQ *A. vinelandii*, Anf clade, 1.55 Å

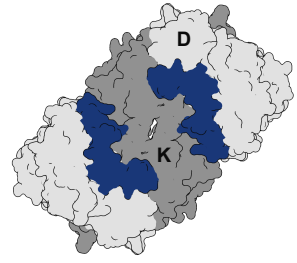


D D-subunit C-terminal extension

52 residues vs. 3U7Q chain D (34 vs. 4WES chain D)

Paper says about 60

② 3U7Q *A. vinelandii*, Nif-I clade, 1.00 Å

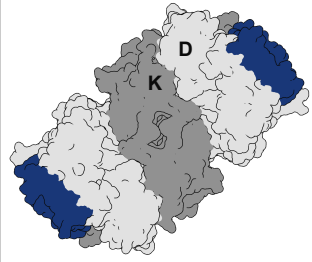


K K-subunit N-terminal extension

43 residues vs. 4WES chain K

Paper says about 8 to 40

③ 4WES *C. pasteurianum*, Nif-II clade, 1.08 Å



D D-subunit insertion

52 residues vs. 3U7Q chain D

Paper says about 55

c Median of each trait, by clade

Today: living proteins; ancestors: reconstructed. Medians over an even count can end in .5.
The 5 deepest ancestors have no clade label and are left out.

	Proteins		Surface area (Å²)		Contacts		ΔG (kcal/mol)	
	today	ancestors	today	ancestors	today	ancestors	today	ancestors
Nif-I	139	137	63,132	63,030	1,191	1,199	−91.6	−92.5
Nif-II	181	179	65,529	66,229	928	935	−77.8	−78.2
Nif-III	31	31	61,753	60,891	892	798	−66.1	−62.3
Vnf + Anf	34	32	59,720	60,164	983.5	991.5	−74.8	−75.0

About the numbers

Paper: Nif-I has the most contacts and the lowest ΔG; Nif-II the largest surface area. The order of the medians matches. Node 802, last common ancestor of Vnf and Anf, is labelled Nif-III in the data.

After Figs 3 and 4 of Cuevas Zuviría et al., eLife 2025, doi:10.7554/eLife.105613, CC BY 4.0. Data: Fig. 4 source data, PDB.
Redesigned; tree and trait values from the source data, medians and segment lengths computed here.